

Danielle Bartz
28-3004 Andrade Rd.
Pepe`ekeo, HI 96783
(954)-643-6408
dbartz@hawaii.edu (school)
daniellebartz3@gmail.com (personal)

Education

August 2019-Current: PhD in Marine Biology, University of Hawai'i at Mānoa, Specialization in Ecology, Evolution, and Conservation Biology (EECB) (Current GPA: 4.0)

May 2018: Bachelor of Science in Geoscience, *Cum Laude*; Minor in Biology, *Biology Honors Distinction*, Florida Atlantic University (GPA: 3.589)

May 2018: Advanced Certificate in Geographic Information Systems (GIS), Florida Atlantic University

May 2018: Certificate in Environmental Science, Florida Atlantic University

May 2014: Associate of Arts, Broward College (GPA: 3.353)

Research Interests

Research interests include marine biology and ecology, conservation biology, elasmobranch migration, habitat suitability, electrophysiology, niche partitioning, GIS, anthropogenic influences on marine ecosystems, and environmental services including the management of invasive or endangered species. I am also interested in Hawaiian culture and the part that local ecological knowledge can play in the reconstruction of past conditions through oral history collection and analysis.

Research Experience

Research Assistant & Hawai'i Island Liaison, Summer 2022-Present
Hawai'i Community Tagging Program, Advised by Melanie Hutchinson, Ph.D.

- ❖ Duties include liaising with community partners, training new fishers on safe offshore tagging procedures, distributing materials to participants, aiding with dive operations to service acoustic receivers, and contributing to the website, quarterly newsletter, and science communications via social media platforms.

Research Assistant, Spring 2022-Summer 2022
Plecostomus/Streamflow Project, HIMB, PI-AK CASC, Advised by Tim Grabowski Ph.D. and Megan Donahue Ph.D.

- ❖ Aided in the collection of invasive *Hypostomus plecostomus* from the Ala Wai Watershed on O'ahu, completed 253 otolith dissections, presented to collaborators in Juneau, Alaska

Volunteer, Fall 2019-Spring 2020

SharkLab, Hawai'i Institute of Marine Biology, Advised by Carl Meyer, Ph.D.

- ❖ Using long-lining, acoustic telemetry and various tagging techniques to assist graduate students in collecting olfactory, reproduction, and migration data in Kane'ohe Bay, O'ahu, HI.

Volunteer, Spring 2020

Community-Based Research Project, Northeast Kauai, Advised by Mehana Blaich Vaughan

- ❖ Traveled to Kauai to interview community members and victims of 2018 floods in Northeast Kauai as a part of ongoing research project. Transcribed and coded 40+ hours of recorded interviews and presented findings at the Waipa Foundation.

United States Coast Guard Auxiliarist AP, Fall 2018-Present

Flotilla 140-03-19 Hilo, Hawai'i

- ❖ Involved in monthly meetings to discuss Flotilla agendas & events, enrolling in specialty training opportunities

Internship/Graduate Research, Summer 2018-Present

Hawai'i Cooperative Fishery Research Unit (HCFRU), Advised by Timothy Grabowski, Ph.D.

- ❖ Collecting water samples for environmental DNA analyses
- ❖ Boating and diving operations for relevant projects
- ❖ Experience dissecting and processing fish otoliths
- ❖ Experience using statistical analyses
- ❖ Community interviews and fishing (CPUE) efforts for graduate research project (2020-2022)
- ❖ Preliminary data collection and establishment of liaisons for my graduate research (2018-2019), examining spatial and temporal dynamics of the Hilo Bay in regards to nursery ground habitat suitability for the Scalloped Hammerhead Shark (*Sphyrna lewini*)
- ❖ Helping lab mate with Tiger Shark Photo-ID project
- ❖ Volunteering in various lab and DAR events such as the otolith dissection station at DAR's tilapia and roi roundups

Scientific Dive Operation with USGS, May 2018

- ❖ Worked with a group of scientific divers coring coral and taking peat samples off the coast of Rose Island, Bahamas aboard the Florida Institute of Oceanography's R/V W.T. Hogarth

Internship, Fall 2017 – Spring 2018

FAU Elasmobranch Research Laboratory, Advised by Stephen Kajiura, Ph.D.

- ❖ Examined the dispersal of electrical fields at various water depths in a lab setting

- ❖ Deployment and retrieval of acoustic telemetry receivers
- ❖ Blacktip Shark Migration Project
- ❖ Shark tagging, sampling, data collection
- ❖ Gained experience deploying and retrieving long lines, trawl nets, and proper shark handling, measuring, and tagging procedures
- ❖ Conducted aerial surveys along coastline
- ❖ Developed dive and float plans for research trips

Honors Thesis Research in Marine Ecology, Summer 2016 – Spring 2017

FAU Marine Ecology Laboratory, Advised by W.R. Brooks, Ph.D. and Chelsea Bennice, M.Sc.

- ❖ **Honors Thesis:** “Diet and Temporal Partitioning by the Common and the Atlantic Longarm Octopuses in a South Florida Habitat”
- ❖ Analyzed video footage to determine the foraging times of two sympatric octopus species
- ❖ Collected and analyzed octopus prey remains using non-invasive field techniques to identify prey species

Directed Independent Study in Marine Ecology, Spring 2016

Wild Dolphin Project, Jupiter, FL, Advised by Denise Herzing, Ph.D. and Sommer Kuhn, M.Sc.

- ❖ Photo identification of bottlenose dolphins from Great Bahama Bank
- ❖ Identified 58 new individuals to be added to the project database

Relevant Coursework (Current GPA: 4.0)

- ❖ BaseR* (Spring 2021)
- ❖ Pono Science (Fall 2020)
- ❖ Communicating Science to Society (Fall 2020)
- ❖ Topics in Marine Conservation (Spring 2020)
- ❖ Care & Collaborative Management of Natural Resources (Spring 2020)
- ❖ Current Research in Marine Biology (Spring 2020)
- ❖ Evoluncheon Seminar (Spring 2020-current)
- ❖ Life Sciences Seminar (Spring 2020)
- ❖ Marine Microplankton Ecology (Fall 2019)
- ❖ Intro to Quantitative Methods (Fall 2019)
- ❖ Life Sciences Seminar (Fall 2019)
- ❖ Career Skills for Marine Biology (Fall 2019)

*Taken through Utah State University, not for credit

Relevant Coursework (Undergraduate)

- ❖ Biology Honors Thesis Program (Intro to Honors I, II, and Honors Research courses) (2016-2018)
- ❖ Spatial Data Analysis (Spring 2018)

- ❖ Biogeography (Spring 2018)
- ❖ Investigations in GIS (Spring 2018)
- ❖ Photogrammetry and Aerial Photo Interpretation (Spring 2018)
- ❖ Applications in GIS (Spring 2018)
- ❖ Geomorphology (Spring 2018)
- ❖ Professional Internship in FAU Elasmolab (Spring 2018)
- ❖ Practical Cell Neuroscience (Spring 2018)
- ❖ Critical Thinking in Environmental Science (Fall 2017)
- ❖ Water Resources (Fall 2017)
- ❖ Urban Geography (Fall 2017)
- ❖ Digital Image Analysis (Fall 2017)
- ❖ Professional Internship FAU Elasmolab (Fall 2017)
- ❖ Intro to Physical Geography (Summer 2017)
- ❖ Remote Sensing of Environment (Summer 2017)
- ❖ Coastal and Marine Science (Summer 2017)
- ❖ Weather and Climate (Summer 2017)
- ❖ Principles of GIS (Spring 2017)
- ❖ Principles of Ecology (Spring 2017)
- ❖ Experimental Design and Statistical Inference (Spring 2017)
- ❖ Introductory Statistics (Spring 2017)
- ❖ Genetics (Fall 2016)
- ❖ Organic Chemistry 1 (Fall 2016)
- ❖ Issues in Human Ecology (Fall 2016)
- ❖ Octopus Research DIS (Summer 2016)
- ❖ General Chemistry 2 & Lab (Spring 2016)
- ❖ Dolphin Identification DIS (Spring 2016)
- ❖ Human-Environmental Interactions in South FL (Spring 2016)
- ❖ Intro to Mapping and GIS (Spring 2016)
- ❖ Cultural Anthropology (Fall 2015)
- ❖ Biodiversity & Lab (Fall 2015)
- ❖ Physical Science (Fall 2015)

Professional Licenses and Certifications

- ❖ Motorboat Operator Certification (MOCC; 2018)
- ❖ AAUS Scientific Diver Certification (2016)
- ❖ DAN Oxygen Administration, First Aid, & CPR (2016)
- ❖ Institutional Animal Care and Use Committee Certifications (2016)
- ❖ Laboratory Safety and Hazardous Waste Awareness & Handling (2016)
- ❖ Biological Safety; Biomedical Waste FAC 64E-16 (2016)
- ❖ PADI Open Water SCUBA Certification

Scientific Presentations

“Influence of streamflow variability on fish populations as a proxy for understanding nutrient transport under climate change in Hawaiian and Alaskan streams”, in-person presentation, US Forest Service/University of Alaska Southeast, PI-AK CASC collaboration (7/25/22)

“Hawai'i's Hammerheads: Understanding Distribution, Nursery Habitats, and Population Trends Over Time”, virtual presentation, Co-op meeting with DLNR Division of Aquatic Resources (5/6/22)

“Hawai'i's Hammerheads: Understanding Distribution, Nursery Habitats, and Population Trends Over Time”, virtual presentation, Tropical Conservation Biology and Environmental Science Symposium 2022, University of Hawai'i at Hilo (4/28/22)

“Hawai'i's Hammerheads: Understanding Distribution, Nursery Habitats, and Population Trends Over Time” virtual presentation, Tester Symposium 2022, University of Hawai'i at Mānoa (4/21/22)

“Working Across Multiple Knowledge Systems”, panelist, HIMB (8/10/21)

“Hilo Bay Hammerheads; Memory Meets Telemetry”, virtual presentation, Honolulu Nerd Nite (6/1/21)

“History of Hammerheads in Hilo: An Interdisciplinary Approach”, virtual presentation, Tester Symposium 2021, University of Hawai'i at Mānoa (4/23/21)

“History of Hammerheads in Hilo: An Interdisciplinary Approach”, virtual presentation, Tropical Conservation Biology and Environmental Science Symposium 2021, University of Hawai'i at Hilo (4/16/21)

“History of Hammerheads in Hilo: An Interdisciplinary Approach”, virtual presentation, EECB, University of Hawai'i at Mānoa (2/5/21)

“History of Hammerheads in Hilo: An Interdisciplinary Approach”, Rapid-Fire Talk, virtual presentation, MBGP, University of Hawai'i at Mānoa (5/8/20)

“Diet and Temporal Partitioning by the Common Octopus and the Atlantic Longarm Octopus in a South Florida Habitat”, poster presentation, Florida Undergraduate Research Conference (FURC) at Florida Atlantic University, Boca Raton, FL. (2/25/2017)

“Diet and Temporal Partitioning by the Common Octopus and the Atlantic Longarm Octopus in a South Florida Habitat”, poster presentation, Undergraduate Research Symposium at Florida Atlantic University, Boca Raton, FL. (3/31/2017)

“Diet and Temporal Partitioning by the Common Octopus and the Atlantic Longarm Octopus in a South Florida Habitat”, poster presentation, Life Sciences South Florida STEM Undergraduate Research Conference at Palm Beach State College, Boca Raton, FL. (4/1/2017)

“Diet and Temporal Partitioning Facilitates Coexistence Between Two Octopus Species in an Intracoastal Waterway”, poster presentation, Life Sciences South Florida Cocktail Reception at FIU Miami Urban Design Studios, South Beach, FL. (6/12/2017)

Publications

Co-authorship; excerpt in *The Value of Hawai'i: 3: Huluhia, the Turning*, 2020, pg. 66-71

Awards

- ❖ **UH Mānoa Marine Biology Graduate Program Student Leadership Award** (Spring 2020)
- ❖ **First Place poster presentation**, OURI Undergraduate Research Symposium at Florida Atlantic University, Boca Raton, FL. (3/31/2017)
- ❖ **First Place poster presentation**, Life Sciences South Florida STEM Undergraduate Research Conference at Palm Beach State College, Boca Raton, FL. (4/1/2017)
- ❖ **Invitational Poster presentation**, Life Sciences South Florida Cocktail Reception at FIU Miami Urban Design Studios, South Beach, FL. (6/12/2017)

Features

- ❖ **GWISH Member Spotlight**, Instagram and website post, June 2021
- ❖ **May 2018: Internship Spotlight: Tracking Blacktip Sharks**, Video released on Official FAU Facebook Page
<https://www.facebook.com/FloridaAtlantic/videos/10156471236012622/>
- ❖ **March 10, 2017: Student Spotlight**, Office of Undergraduate Research & Inquiry, FAU http://www.fau.edu/ouri/student_spotlight.php
- ❖ **March 17, 2017: How did you luck into getting involved in research?** Office of Undergraduate Research & Inquiry, FAU
- ❖ **May 15, 2017: “Things I wish I knew as an undergraduate researcher”**, Panelist at workshop held by the Office of Undergraduate Research & Inquiry
- ❖ **June 28, 2017: Student Spotlight**, Office of Undergraduate Research & Inquiry, FAU

Teaching Experience:

- ❖ Fall 2021: BIOL 171L (3 sections online). School of Life Sciences, University of Hawai'i at Mānoa
- ❖ Spring 2021: BIOL 171L (3 sections online). School of Life Sciences, University of Hawai'i at Mānoa

- ❖ Fall 2020: BIOL 171L (3 sections online). School of Life Sciences, University of Hawai'i at Mānoa
- ❖ Spring 2020: BIOL 171L (2 sections). School of Life Sciences, University of Hawai'i at Mānoa
- ❖ Fall 2019: BIOL 171L (2 sections). School of Life Sciences, University of Hawai'i at Mānoa

Leadership

- ❖ **School of Life Sciences New Student Orientation**, host & panelist (Summer 2020 & 2021)
- ❖ **UH Mānoa Marine Biology Graduate Program Student Leadership Award** (Spring 2020)
- ❖ **iLead 10th Annual Student Leadership Conference**: Worked closely with the Weppner Center for LEAD & Service-Learning to host a breakout session at the iLead Conference on 9/23/17 titled "I am Real". In this session my co-host, Madeline Rubio and I used an activity called "Conscious Decisions" to define "real", discuss values, morals, and ethics, and then prompt participants with scenarios and see if their decisions aligned with their values. We had a total of 75 participants in our activity for this breakout session.
- ❖ **Graduate of the LeaderShape Institute** (2017)

Academic Organizations

- ❖ **2020 – Current**: Graduate Women in Science Hawai'i (GWISH)
- ❖ **Fall 2019 – Fall 2021**: Graduate Representative for the Marine Biology Graduate Program, UH Mānoa
- ❖ **2016 - 2018**: President of SEEDs (Strategies for Ecology Education, Diversity, and Sustainability) Ecology Club, the FAU student chapter of the ESA. As President it was my duty to send out emails to our 230+ student members, organize and host the biweekly meetings, take charge of promoting the club as well as fundraising events, and lead our Trail Maintenance of the FAU Preserve. Events included:
 - Hosted bi-weekly General Body Meetings with student members to discuss organization business (Spring & Fall semesters).
 - Hosted monthly "Trail Maintenances", bringing groups of students (ranging 5-15) to clean up garbage and clear brush in the 90-acre FAU Preserve (Spring & Fall semesters).
 - "1st Annual Holiday Hoot" event at the FAU Preserve with a total of 137 volunteers & attendees. This event showcased the importance of ecological science and research while having activities like ornament crafting, cookie decorating, lighted walks through the FAU Preserve, ecology corner, pictures with the school mascot, free raffles, and hot chocolate (11/30/16).

- Tabling event representing SEEDs at Science Meeting of the Minds (10/25/16)
- Tabling event representing SEEDs at College Connect event (9/28/16)
- Organized & hosted Invasive Species Workshop with faculty guest speaker Diane Owen, PhD (9/28/16)
- Tabling event representing SEEDs at ClubFest (9/13/16).
- Movie night (4/11/17, 4/20/17)
- Fun run (4/23/17)
- Earth Day tabling event with Mission Green (4/20/17)
- “2nd Annual Holiday Hoot” event in Biological Sciences (Sanson Science Building, FAU) patio with a total of 55 volunteers & attendees. This year I incorporated other academic organizations by having them table at the event and give out information about their clubs. This event showcased the importance of ecological science and research while having activities like ornament crafting, ecology/info corner, pictures with the school mascot, free raffles, and pizza all engulfed within a plethora of holiday lights and decorations. (11/13/17).
- ❖ **2017 - 2018:** Outreach Director for the Council for Scholarship & Inquiry (CSI). As Outreach Director it was my job to help get students involved in research through various channels. I worked directly with the Office of Undergraduate Research and Inquiry to reach out to undergraduates and help open their eyes to all of the opportunities available in the realm of research.
 - Planned and hosted Science Meeting of the Minds (MOM) event in the College of Biomedical Sciences, where the organization hosted a panel of professors speaking on behalf of research opportunities within their labs. (10/23/17)
 - Planned and hosted Engineering Meeting of the Minds (MOM) event in the College of Engineering, where the organization hosted a panel of professors speaking on behalf of research opportunities within their labs. (11/14/17)
- ❖ **2017 - 2018:** Peer Mentor for the Office of Undergraduate Research & Inquiry (OURI). As a Peer Mentor I assisted other undergraduates with getting involved in research, writing and editing papers/posters, and provided mentorship over their individual projects. I held weekly office hours and gave numerous classroom presentations to large lecture classes on behalf of Peer Mentors and the Office of Undergraduate Research & Inquiry. I also hosted a number of workshops throughout the semester to help students learn about the research opportunities on campus. Workshops and presentations included:
 - “Applying for Grad School” Workshop (2/6/18)
 - FAU High School Research Presentation (8/28/17)
 - “How to Apply for Grants” Workshops (9/21/17, 10/2/17)
 - “Attending a Conference” Workshop (10/25/17)
 - “Creating an Oral and Poster Presentation” Workshop (11/6/17)
 - “How to Get Started” Workshop (11/14/17)
 - FAU High School Research Presentation (1/22/18)
- ❖ **2017 - 2018:** Solar Energy Society, member
- ❖ **2017 - 2018:** Visionary Owls, member

Memberships

- ❖ American Fisheries Society (AFS)
- ❖ USCG Auxiliary Flotilla 140-03-19 (CGAUX)
- ❖ Graduate Women in Science Hawai'i (GWISH)
- ❖ American Elasmobranch Society (AES)
- ❖ American Academy of Underwater Sciences (AAUS)
- ❖ Council for Undergraduate Research (CUR)
- ❖ Student Chapter of the Ecological Society of America (SEEDS-ESA)
- ❖ Divers Alert Network (DAN)

Grants & Scholarships

Spring 2020: UHM MBGP Student Leadership Award (\$600)
Spring 2020: MBGP Achievement Scholarship (\$500)
Fall 2019: Kūlana Noi'i Award for Biocultural Community Engagement (\$1000)
Spring 2018: James P. and Eloise M. Latham Scholarship in Geography (\$1000)
Spring 2018: Research Funding through OURI (\$300)
Fall 2017: Research Funding through OURI (300)
Spring 2017: OURI Undergraduate Research Grant (Brooks Lab) (\$600)
Spring 2016: Florida Medallion Scholarship (\$1001)
Fall 2015: Florida Medallion Scholarship (\$1001)

Nominations

Spring 2022: Frances Davis Award for Excellence in Undergraduate Teaching
Spring 2021: UH Mānoa School of Life Sciences Outstanding Teaching Assistant Award
Spring 2021: Frances Davis Award for Excellence in Undergraduate Teaching
Spring 2020: UHM MBGP Student Leadership Award (accepted)
Spring 2020: Frances Davis Award for Excellence in Undergraduate Teaching
Spring 2018: Florida Atlantic University "Luncheon with President Kelly to Recognize Excellence in Research" (accepted)
Spring 2018: Dean's Scholarship Luncheon (accepted)
Spring 2018: Florida Atlantic University 2018 Undergraduate Researcher of the Year
Spring 2018: Florida Association of Colleges & Employers 2018 Student of the Year Award
Spring 2017: The Society for Collegiate Leadership & Achievement

Technical and Specialized Skills

- ❖ Proficient in fish dissection, otolith collection (pulling, polishing, mounting, reading)
- ❖ Intermediate competency in Geoscience Information System (GIS), (ArcMap, ArcCatalog, ArcPad, ESRI, ERDAS Imagine, ENVI, Google Earth, TigerWeb, Trimble)

- ❖ Proficient use of multiple electric field recording instruments including AD boards, HumBug Noise Eliminators, various amplifiers, as well as building electrodes.
- ❖ Intermediate competency in Neurons in Action (NIA) neuron simulation software
- ❖ Intermediate competency in LabChart Software
- ❖ Proficient in use of DinoXcope Software
- ❖ Microsoft Office (Word, Powerpoint, Excel)
- ❖ Intermediate speaking, writing, and comprehension of the Spanish language

Outreach, Extracurricular Activities and Services

- ❖ Volunteer at Aloha 'Aina quarterly community cleanup events in Makahanaloa Ahupua'a, 2020-current
- ❖ Volunteer at SeaHorse Stables, 2018-current, helping with equestrian training, exercise, medical care, and barn/pasture upkeep
- ❖ Volunteer at Waimanalo Limu Hui, 2020
- ❖ Volunteer at Waipa Lo'i Kalo Community Work Day, 2020
- ❖ Volunteer at Keaukaha Fishpond Restoration Community Work Day, 2019
- ❖ Volunteer at Gumbo Limbo Nature Center Spring 2018
- ❖ Volunteer removing invasive plant species (i.e. *Epipremnum aureum*, commonly known as Golden pothos vines) at various local parks, Spring 2018
- ❖ Volunteered for roughly 5 years at Abandoned Pet Rescue, a shelter where I would tend to many different animals, ranging from cats with AIDS and Leukemia to 4-foot long iguanas. I also fostered and rescued 3 pets.
- ❖ Participated in numerous fundraisers and beach clean-ups while in Girl Scouts for 12+ years